

Foundation class for 100xdevs cohort

What we're learning

1. VSCode
2. HTML
3. CSS
4. JS
5. DOM
6. Assignments

What we'll learn practically

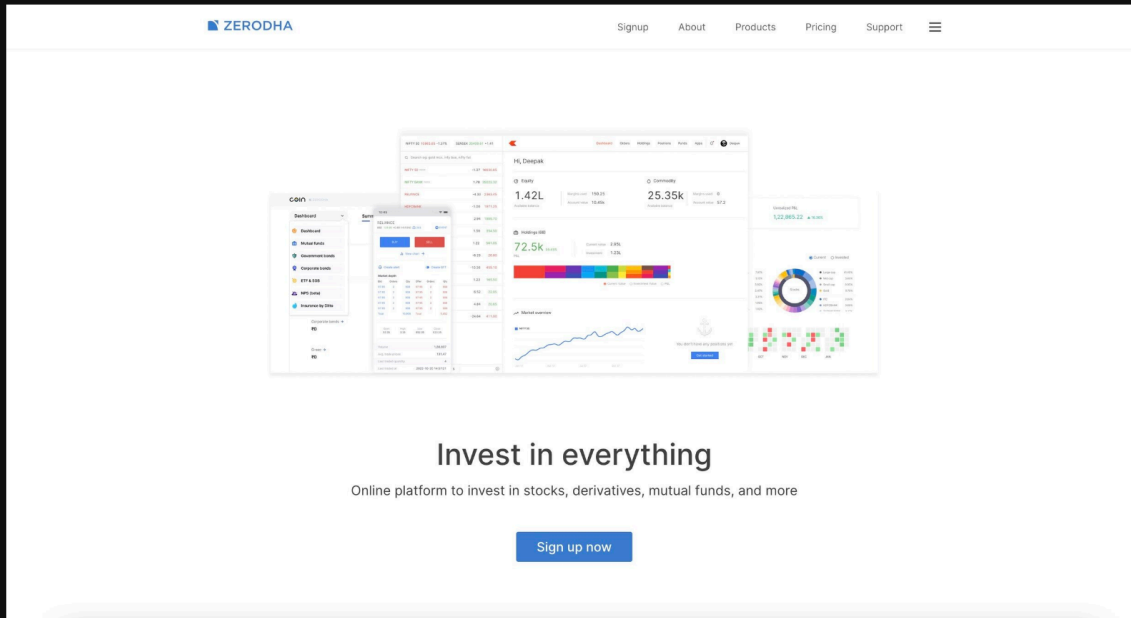
Small boi stuff

1. Tags
2. Attributes
3. Inline styles (bg color, fonts ...)
4. JS Basics, data types, loops, fns...

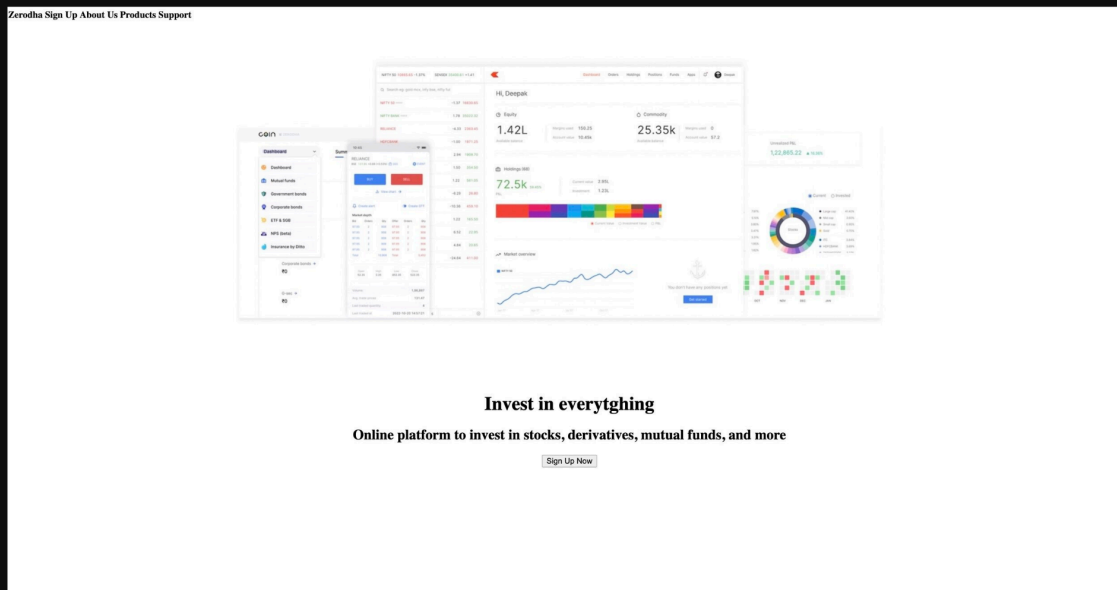
Big boi stuff

1. External styles, classes and ids
2. Flexbox
3. Connecting JS to HTML (selectors, DOM)
4. Chrome Dev tools

What we're building

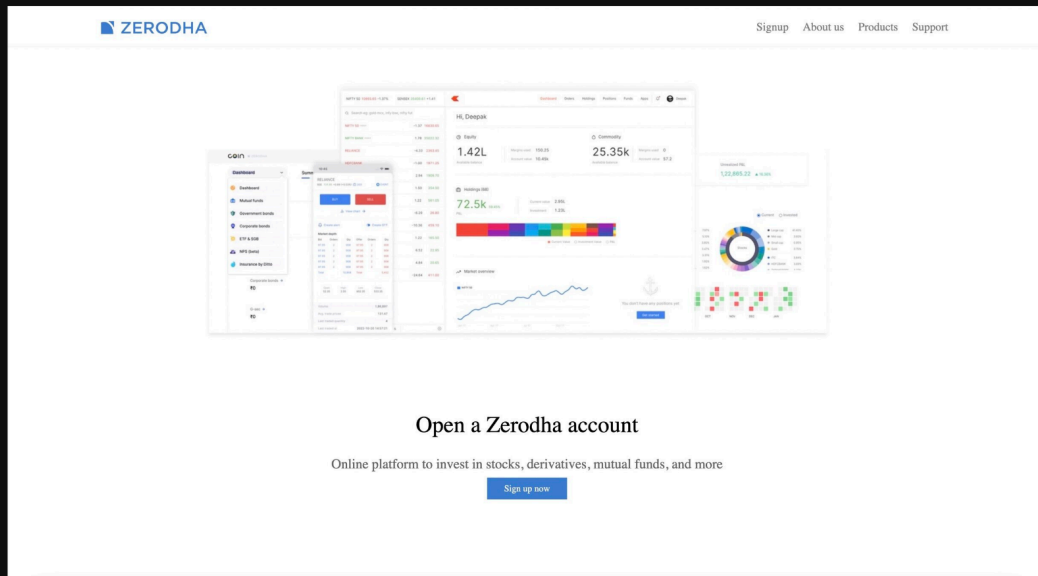


Stage 1 (HTML Only)



Stage 2

(HTML + CSS)



Stage 3

JS

Create a portfolio returns calculator

A hand-drawn sketch of a portfolio returns calculator interface. It features a rounded rectangular container with four input fields on the left, stacked vertically: "Capital", "Years", "Interest", and a wide "Calculate" button at the bottom. To the right of the "Capital" field is a separate rounded rectangular box containing the text "4 Cr".

Before we start

What are browsers? How do they **render** websites?

What purpose do HTML/CSS/JS serve

How much of this do we actually use in the industry -

1. html => very minimal

2. css => minimal

3. js - heavy

IDE

VSCode

Vim

WebStorm

HTML

Defines the structure of your website

2 jargon we need to know of -

1. Tags
2. Attributes

HTML Tags

- `<html>`
- `<head>`
- `<title>`
- `<body>`
- `<div>` / `<span>`
- `<h1>` .. `<h6>`
- `<p>`
- `<img>`
- `<a>`
- `input`
- `<button>`
- `<b>` / `<i>`
- `<center>`

Attributes

- ``
 - `<a href="google.com">`
 - `<button onclick="">`
 - `<input id="username">`
-
- `<input id="username">`

Css

They let you add **styles** to your website (colors, font sizes, background colors ...)

Used for positioning things on the page

How to add styles?

```
3  
4  <h1>hi there</h1>
```

File | /Users/kirat/Projects/100xdevs/foundation-class/index.html

hi there

How to add styles?

```
3  
4 <h1 style="color: red; background: green;">hi there</h1>
```

File | /Users/kirat/Projects/100xdevs/foundation-class/index.html

hi there

Common styling attributes

- color
- background
- border-radius
- border
- padding / margin
- box-shadow

How to position?



divs always take up all the space available horizontally
spans only take up as much space as needed

Can you make divs take only the space they need?

How to position?



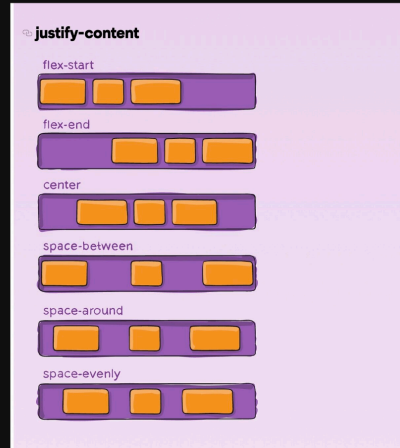
Dumb way



float property

Right way Flex box

display: flex
on parent
makes all div siblings reside
in the same line



Javascript

It is a scripting language

It can be used to solve algorithmic problems/do ML/do
backend development

Specifically for websites, it is useful to add functionality to
your website (for example, a returns calculator)

We're going to practise on repl.it first

For JS (or any language), you need to know

- variables
- Simple Data types (strings, numbers, boolean)
- Complex Data types (arrays and objects)
- Control statements and operators
- Loops
- Functions

Problems to understand these

1. Count from 1 to 1000
2. Is Palindrome?
3. Find the name of the oldest person in a group

Take home assignment -

Find the age of k oldest people in the group

<https://leetcode.com/problem-list/top-interview-questions/?topicSlugs=string&page=1&difficulty=EASY>

<https://leetcode.com/problem-list/top-interview-questions/?topicSlugs=array&page=1&difficulty=EASY>

Connecting JS to returns tracking app

First let's write a function that takes 3 inputs

1. numberOfYears
2. base
3. yearlyGains

and returns net portfolio value

Assignment

A hand-drawn calculator interface with a rounded rectangular border. It contains the following elements:

- Four input fields stacked vertically on the left: "Capital", "Years", "Interest", and a wide "calculate" button at the bottom.
- A checkbox labeled "Compound?" in red text, located to the right of the "Capital" field.
- A single input field containing the text "4 Cr", located to the right of the "Years" field.